INCH-POUND

MIL-PRF-49471/2(ER)
2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X598/U

This specification is approved for use within Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49471(ER).

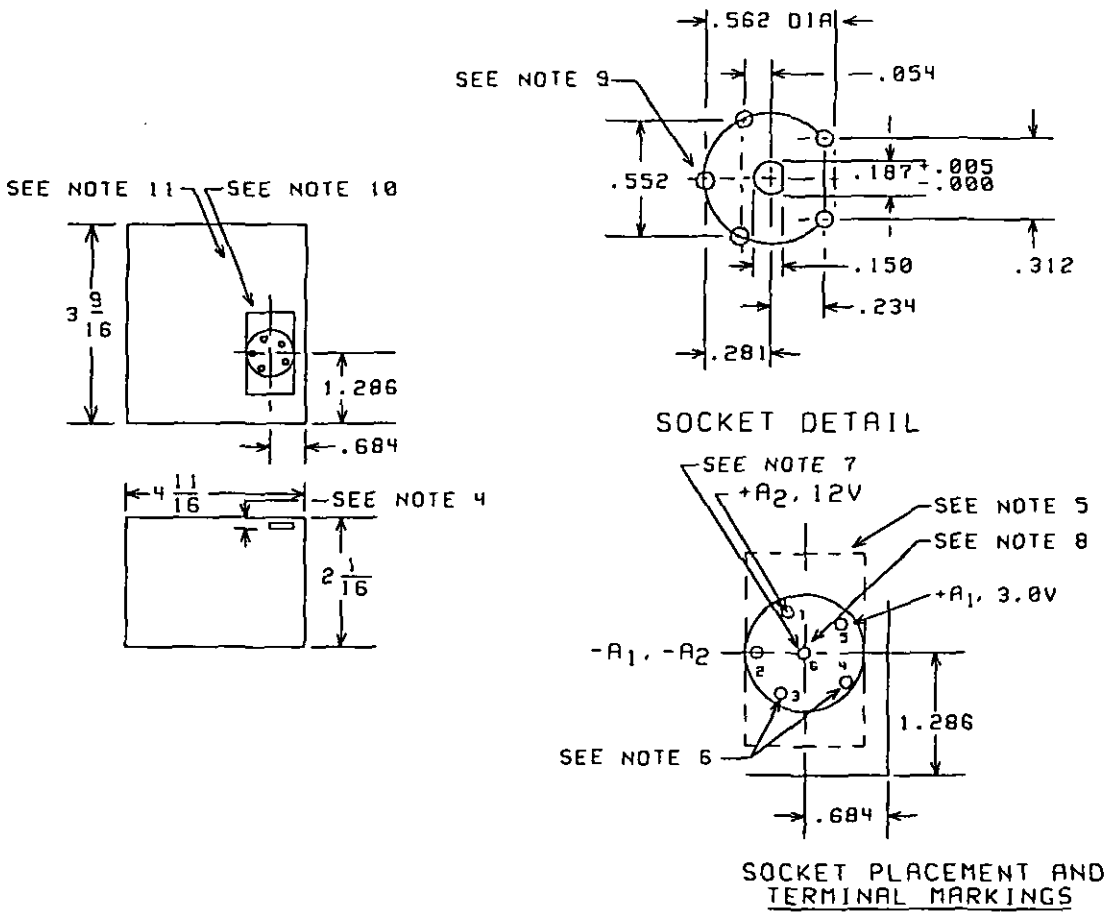


Figure 1. Battery dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are ±0.005 inch for decimals.

3. Tolerances for length, width, and height of battery are ±1/16 inch for fractions.

4. The minimum depth of socket well shall be 1/2 inch.

5. The size of the insulating plate is optional.

6. Socket insert contact not required.

7. Pin-circle center.

8. "D" hole for plug positioning.

9. Five holes to fit 1/8 inch pins, 7/16 inch in length.

10. Socket must float 3/32 + 1/32 -1/16 inch in all directions from the specified pin circle center location in the plane of the socket from reference dimension.

11. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during and after subsequent insertions of the mating plug during contractual testing.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

	A_1 Unit	A_2 Unit
Maximum voltage:	3.05	15.25 volts
Minimum voltage:	2.00	10.00 volts

Charge protection device: Applicable.

Safety feature: Applicable.

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Terminals: Five (5) hole socket type with no obstruction of any holes.

Weight (maximum): 1.5 pounds (681 grams).

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 10.00 Volts
I	3.8
L	2.0
H	6.5
HT	6.3
LT	1.9
IT	3.25

Initial voltage delay: When the battery is subjected to the I, H, HT, and IT capacity tests, the time required at the beginning of discharge of the battery to reach a voltage of 10.00 volts after the specified load is applied shall not be more than 5.0 seconds. For the L and LT capacity tests the time to reach a voltage of 10.00 volts after the load is applied shall not be more than 60 seconds.

High temperature device: Applicable.

Complete discharge device: Applicable.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for capacity tests are specified in basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 2.0 amperes at 70±5°F to 2/3 of open-circuit voltage. Each discharged cell shall then be connected in series to four fresh cells. Each string shall be discharged at 2.0 amperes at 70±5°F to 2/3 open-circuit voltage of the four fresh cells.

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Batteries: The A_2 unit ($-A_1$, $-A_2$, to $+A_2$) shall be discharged as follows: For all capacity tests, except the H and HT, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

Closed-circuit voltage: Closed-circuit voltage measurements shall be made with a direct current voltmeter of proper range and sensitivity (see basic specification), using load resistance as shown in the following table:

UNIT RESISTANCE	MINIMUM PERMISSIBLE VOLTAGE
A_1 6.5 ohms, $\pm$ 1%	1.9 volts
A_2 14.2 ohms, $\pm$ 1%	10.0 volts

Voltage shall be above "MINIMUM PERMISSIBLE VOLTAGE" within five (5) seconds after load is applied.

Socket strength test (First Article only): Before conducting the complete discharge test, the five batteries shall be subjected to the following:

Using a 0.250 $\pm$0.050 inch diameter rod, apply a force to the center of the socket and deflect it downward a distance of 0.250 +0.0625, -0 inch below the top surface of the battery.

Battery Label (only for this battery):

"DO NOT CHARGE, SHORT CIRCUIT, INCINERATE, MUTILATE OR TEST WITH U-410 ADAPTER OTHERWISE BATTERY MAY VENT OR RUPTURE RELEASING TOXIC MATERIALS"

Custodian:
Army-ER

Preparing Activity:
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(Project 6135-A274-02)

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